BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Operating Systems: Interfaces and Memory

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. After marking an assessment, name one error category you should identify.

2. Which is malware?
   A) Ransomware
   B) A trusted security update
   C) A spreadsheet formula
   D) An authorised backup

3. Given A has code 65 and capital-letter codes are consecutive, give the code for C and explain how you found it.

4. A website needs English, Greek and Chinese text. "Unicode has fewer characters than standard ASCII." Correct this. Why does Unicode suit this site?

Today's objectives

Describe how an operating system provides a user interface. Explain allocation of memory and how an OS supports multitasking.

Key vocabulary - Read, cover, recall

Operating system
System software managing hardware/resources and supplying services to programs.

GUI
Graphical User Interface using windows, icons, menus and a pointer.

CLI
Command Line Interface using text commands.

Memory management
Allocation and management of RAM for running programs.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Why an operating system?

System software coordinates hardware and applications, and provides services such as a user interface, memory management and security. Without it, applications would have to manage many hardware details.

Graphical and command interfaces

A GUI lets users select icons, windows and menus, making common tasks approachable. A CLI accepts typed commands, which can be efficient for an experienced user and automation.

Allocating memory

The OS allocates available RAM to applications, manages which data is kept in RAM and moves data as needed. Paging and segmentation are not required for this specification.

Multitasking

The OS manages running applications and shares processor time; it allocates memory and handles resources so users can switch between tasks. Switching does not mean one CPU core necessarily runs every task at exactly the same instant.

I Do - Worked model

A student has a browser, music app and editor open. Explain how the OS manages their memory and allows multitasking. [4 marks]

- Identify the three applications competing for resources.
- The OS allocates space in RAM to applications [1], keeping their required data available [1].
- The OS manages CPU time among tasks [1], so the student can switch between or use different applications [1].
- Four linked marks: memory allocation, purpose, processor time, multitasking.

We Do - Guided practice

1. A receptionist runs booking software and an email client together. Explain TWO ways the OS helps these applications operate. [4 marks]

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2. A technician prefers typed commands for repetitive setup tasks. Explain one benefit and one drawback of a CLI compared with a GUI. [4 marks]

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3. A computer has three programs open but limited RAM. Explain TWO memory-management problems the OS must handle. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict free RAM after loading the application (MB).

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

You Do - Independent exam questions

Q2. State TWO functions of an operating system. [2 marks]

Q3. A photographer switches between photo-editing software and a browser. Explain how operating-system memory management and multitasking support this. [4 marks]

Q4. Compare GUI and CLI interfaces for an office worker and explain the OS role in memory management. [6 marks]

Exit Ticket - True or False?

The operating system can allocate RAM to different applications to help them run.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Missing knowledge, a wrong technical term, a missing cause and effect, or a missing detail.

2. A - Ransomware

3. 67; C is 2 position(s) after A.

4. Unicode supports far more characters than standard ASCII. It can cover English, Greek and Chinese. Use an encoding with enough codes; Unicode characters do not always take one byte.

I Do - Explain the model

Identify the three applications competing for resources.

The OS allocates space in RAM to applications [1], keeping their required data available [1].

The OS manages CPU time among tasks [1], so the student can switch between or use different applications [1].

Four linked marks: memory allocation, purpose, processor time, multitasking.

We Do - Model answers

1. The OS allocates available RAM to each application [1], allowing program data to be held in primary memory [1]. It manages CPU time or resources [1], letting both applications remain usable through multitasking [1].

2. A CLI can run repeated or scripted commands efficiently [1], speeding up frequent technical tasks [1]. Users must remember accurate commands [1], so beginners may find it harder and make errors [1].

3. The OS must allocate the available RAM across programs [1] to prevent applications from competing for the same allocated space [1]. It manages moving/loading application data in memory [1], so programs have needed data as the user switches between them [1].

Code First - Answer

```
freeMB = 1200
appMB = 850
print(freeMB - appMB)
```

Expected output: 350

You Do - Self-mark

Q2: Provides a user interface [1]; allocates/manages RAM for applications [1]. Other valid functions include file, device, user management and multitasking.

Q3: OS allocates RAM to each program [1] so needed data is available [1]. It manages the running tasks and processor time [1], so the user can switch between programs while both remain active [1].

Q4: GUI uses visual windows/icons and menus [1], making routine functions easier to discover [1]. CLI takes typed commands [1] that can be efficient for expert repetitive tasks [1]. OS allocates RAM to programs [1], allowing applications to run with required data in memory [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.